Tertiary Infotech Academy Pte Ltd

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ACTIVITY 3 — TRAINER PACK

For

ISO 14001:2015 Internal Auditor Training

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Conducted by

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Activity 3 — EcoTech Electronics — Risk Assessment, CARs and the Audit Report

Topic 3 · **75 minutes** · Assesses **K4 · A3** · Course TGS-2025054479

Learning purpose

EcoTech failed an NEA inspection for exceeding its copper discharge limit, three hazardous waste containers are unlabelled, the spill bund is cracked, and the EHS Manager post has been vacant for three months. Assess the risks, write the CARs, and draft the report.

Scenario

EcoTech Electronics Pte Ltd is a contract electronics manufacturer assembling printed circuit boards (PCBs) and electronic components for international clients. It operates a 10,000 m² facility in **Tuas, Singapore**, with 200 employees.

EcoTech's operations generate several environmental concerns:

- **hazardous waste** — solder dross, flux residues and waste solvents;
- **wastewater containing heavy metals** (lead, tin, copper) from the PCB washing process;
- **air emissions** from soldering operations.

EcoTech recently **failed a regulatory inspection by the National Environment Agency (NEA)**, receiving a warning for exceeding the permitted trade effluent discharge limit for **copper concentration**. In addition, a routine check by the facilities team found that **three hazardous waste storage containers in the soldering area were unlabelled**, and the **spill containment bund was cracked and leaking**.

The company's risk register was last updated **eight months ago** and does not reflect recent changes:

1. introduction of a **lead-free soldering process** using higher temperatures and different flux chemicals;
2. a **30% increase in production volume** from a new client contract;
3. the **departure of the EHS Manager three months ago** — the position remains vacant.

You have been tasked with conducting a preliminary risk assessment of EcoTech's environmental management practices and preparing an audit report with corrective action recommendations.

The evidence you have been given

- Contract electronics manufacturer, 10,000 m² in Tuas, 200 employees.
- Hazardous waste: solder dross, flux residues, waste solvents (toxic industrial waste).
- Wastewater with lead, tin and copper from PCB washing.
- Air emissions from soldering.
- NEA WARNING — trade effluent copper concentration exceeded the permitted limit.
- 3 hazardous waste containers in the soldering area found UNLABELLED.
- Spill containment bund CRACKED and LEAKING.
- Risk register 8 months old — predates all three changes below.
- Change 1: lead-free soldering — higher temperature, different flux chemistry.
- Change 2: production volume +30% on a new client contract.

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- Change 3: EHS Manager left 3 months ago; post still vacant.

What you will produce

DELIVERABLE — A prioritised environmental risk assessment, three fully-written Corrective Action Requests with root cause analysis, and the skeleton of the audit report to EcoTech's management.

Step-by-step method

Step 1: Separate the aspects from the impacts

Build a register with columns Activity | Aspect | Impact | Condition (Normal / Abnormal / Emergency). Discipline: the ASPECT is the interaction you control ('discharge of trade effluent containing copper'); the IMPACT is the change to the environment ('contamination of the watercourse, toxicity to aquatic life'). Writing 'copper' or 'pollution' as an aspect is the classic error.

Step 2: Cover all three operating conditions

Normal: routine PCB washing discharge, soldering emissions, waste generation. Abnormal: higher-temperature lead-free soldering, the +30% volume surge, cleaning and changeover. Emergency: bund failure releasing solvent to the drain, a fire in the solvent store. The cracked bund is an EMERGENCY-condition aspect with a control that has already failed — that combination is what makes it urgent.

Step 3: Score each risk on the 5×5 matrix

Severity 1 (Insignificant) to 5 (Catastrophic); Likelihood 1 (Rare) to 5 (Almost certain). Score = S × L. Bands: 1–4 Low, 5–9 Medium, 10–14 High (SIGNIFICANT), 15–25 Extreme (SIGNIFICANT, act now). Then apply the OVERRIDE rules — an aspect is automatically significant if it is subject to a legal limit or licence condition, has attracted a regulatory notice in the last 3 years, has generated a substantiated complaint, or involves a controlled hazardous substance or toxic industrial waste.

Step 4: Identify at least five significant risks and rank them

The copper exceedance is automatically significant (legal limit AND a regulatory warning). The cracked bund is automatically significant (emergency aspect, controlled substances, failed control). The unlabelled hazardous waste containers are automatically significant (toxic industrial waste). Then the stale risk register after three material changes, and the vacant EHS Manager role — the competence and resource gap that explains the other four.

Step 5: Run 5 Whys on the copper exceedance

Do not stop at 'the operator did not check'. Push until you reach a SYSTEM cause: the exceedance -> pre-treatment not removing copper -> loading rose after the +30% volume increase -> the pre-treatment capacity was never re-assessed -> no management-of-change step requires an environmental re-assessment when volume rises -> and there is no EHS Manager to own that step. Root cause: the change-management process does not trigger re-evaluation of aspects and controls.

Step 6: Distinguish correction from corrective action

Correction fixes the symptom: re-label the three containers, repair the bund, re-dose the effluent. Corrective action eliminates the cause: amend the management-of-change procedure to require an environmental impact re-assessment for any volume or process change, fill the EHS Manager role, and re-run the aspects evaluation. Clause 10.2 requires the latter — a CAR that only lists corrections is an incomplete CAR.

Step 7: Write each CAR in the three-part structure

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REQUIREMENT (the exact clause or legal obligation breached) + EVIDENCE (what you saw, specific and traceable — location, date, document number, quantity) + NONCONFORMITY (the plain statement of the gap). Then grade it Major or Minor and state why. Add root cause, correction, corrective action, owner and due date.

Step 8: Grade the findings and justify the grade

Major: a legal breach, a total absence of a required process, or a systemic breakdown. The copper exceedance with an NEA warning is Major. The stale risk register after three changes is Major (6.1.2 explicitly requires change to be considered). The unlabelled containers, if isolated, are Minor — but combined with the cracked bund they evidence a systemic failure of operational control under 8.1.

Step 9: Draft the audit report

Sections: (1) audit objective, scope, criteria and dates; (2) audit team and auditees; (3) the sampling basis and its limitations; (4) findings — each graded, with the clause and the evidence; (5) positive observations; (6) opportunities for improvement, shown separately from nonconformities; (7) the CONCLUSION on whether the EMS conforms and is effectively implemented and maintained; (8) agreed corrective action owners and due dates; (9) distribution and follow-up arrangements.

Questions to answer

Q1. Conduct a preliminary risk assessment: identify at least five significant environmental risks, assess each using a likelihood and consequence framework, and prioritise them for audit attention.

What a good answer contains: Look for correct aspect/impact separation, all three operating conditions covered, explicit S×L scores, and — most importantly — the override rules applied so that the legally-breached copper discharge is significant regardless of arithmetic.

Q2. For the three most critical risks, write effective Corrective Action Requests including the nonconformity description, the relevant ISO 14001:2015 clause, the root cause analysis, and specific corrective and preventive actions.

What a good answer contains: Each CAR must have all three parts. Reject 'the operator was careless' as a root cause — it must reach a system or process gap. Correction and corrective action must be clearly distinguished.

Q3. Outline the key sections and content of the audit report to EcoTech's management.

What a good answer contains: Must include the conclusion on effectiveness, the sampling limitation, and OFIs kept separate from nonconformities.

Self-check

YOU ARE DONE WHEN — Your register separates aspect from impact correctly on every line, covers normal / abnormal / emergency conditions, applies the override rule so the copper exceedance is significant, your 5 Whys reaches a process cause (not a person), and each of your three CARs contains requirement + evidence + nonconformity with a stated grade.

If you are stuck — what usually goes wrong

- You wrote 'copper' or 'pollution' in the aspect column. The aspect is the INTERACTION ('discharge of trade effluent containing copper').
- You scored the copper exceedance as medium and left it off the significant list. A breached legal limit is significant regardless of the arithmetic.

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- Your 5 Whys stopped at 'the operator did not check'. Push to the missing process step — that is what corrective action can fix.
- Your CAR lists only corrections (re-label, repair). Clause 10.2 also requires action to eliminate the CAUSE.
- You graded everything Major. Grade against your defined criteria and justify each one; over-grading destroys credibility as fast as under-grading.
- You omitted the sampling statement from the report. It is the most commonly missed element and it protects you.

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Model answer (trainer only)

Five significant risks (ranked).

1. **Trade effluent copper exceedance** — Aspect: discharge of trade effluent containing copper from PCB washing. Impact: watercourse contamination, toxicity to aquatic life. S=4, L=4 → 16 Extreme. Automatically significant: legal limit breached AND an NEA warning issued. Clauses 6.1.3, 8.1, 9.1.2.
2. **Cracked, leaking spill containment bund** — Aspect: potential release of solvents and metal-bearing liquids (emergency condition). Impact: soil and groundwater contamination. S=5, L=3 → 15 Extreme. The designed control has already failed. Clauses 8.1, 8.2.
3. **Three unlabelled hazardous waste containers** — Aspect: storage of toxic industrial waste without identification. Impact: mis-handling, incompatible mixing, improper disposal, contamination. S=4, L=4 → 16 Extreme. Automatically significant (toxic industrial waste). Clauses 8.1, 7.2, 7.3.
4. **Risk/aspects register 8 months stale across three material changes** — the lead-free process changes the flux chemistry and emission profile; +30% volume changes every loading rate. S=4, L=5 → 20 Extreme. Clause 6.1.2 explicitly requires change to be considered.
5. **EHS Manager post vacant for three months** — no owner for aspects, legal register, monitoring or compliance evaluation. S=3, L=5 → 15 Extreme. Clauses 5.3, 7.1, 7.2. This is the resource gap underlying risks 1–4.

CAR 1 — Major. *Requirement:* ISO 14001:2015 clause 6.1.3 requires the organisation to determine its compliance obligations and take them into account in the EMS, and clause 8.1 requires operational controls established against operating criteria; EcoTech's trade effluent discharge is subject to a permitted copper concentration limit. *Evidence:* NEA inspection warning dated [date] recording copper concentration above the permitted limit at the site discharge point; pre-treatment plant records show no capacity re-assessment following the 30% production increase. *Nonconformity:* the organisation has not maintained operational control sufficient to fulfil its compliance obligation for trade effluent copper. *Root cause (5 Whys):* the management-of-change process contains no step requiring environmental re-assessment when production volume or process chemistry changes. *Correction:* re-dose and re-test the effluent; notify NEA of actions taken. *Corrective action:* amend the management-of-change procedure to mandate an environmental impact and capacity re-assessment for any volume or process change; re-assess pre-treatment capacity at the new loading; add continuous monitoring where practicable. *Grade:* MAJOR — a breach of a legal compliance obligation confirmed by a regulator.

CAR 2 — Major. *Requirement:* clause 6.1.2 requires environmental aspects to be determined considering change, including new or modified activities, and reasonably foreseeable emergency situations. *Evidence:* the environmental aspects and risk register is dated [8 months prior] and contains no entry for the lead-free soldering process introduced [date], the 30% volume increase, or the revised flux chemistry. *Nonconformity:* environmental aspects have not been re-determined following material changes to activities and processes. *Root cause:* same change-management gap, compounded by the vacant EHS Manager role leaving the register unowned. *Correction:* re-run the aspects identification for all three changes. *Corrective action:* assign ownership of the register; require aspects review as a gate in the change process; set a periodic review frequency. *Grade:* MAJOR — a systemic failure of a core planning process.

CAR 3 — Minor (escalating to Major if systemic). *Requirement:* clause 8.1 requires operational controls consistent with operating criteria; hazardous waste storage must permit correct identification and handling. *Evidence:* on [date], three waste containers in the soldering area were observed without labels identifying contents or hazards; the adjacent spill containment bund was cracked with visible leakage. *Nonconformity:* operational controls for hazardous waste storage and spill containment are not implemented as required. *Root cause:* no scheduled inspection of the waste storage area and no

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bund integrity testing regime; competence and awareness gaps following the EHS Manager's departure. *Correction:* label the three containers; repair the bund; contain and clean the leak. *Corrective action:* introduce a documented weekly waste-area inspection with a checklist; add bund integrity testing to the maintenance schedule; re-train soldering staff on waste labelling. *Grade:* MINOR individually; raise to MAJOR because it occurs together with CAR 1 and CAR 2, evidencing a systemic operational-control breakdown.

Report sections. Objective, scope, criteria and dates · audit team and auditees · sampling basis and its limitations · findings graded with clause and evidence · positive observations · opportunities for improvement (kept separate from nonconformities) · the conclusion on conformity and effective implementation · agreed corrective action owners and due dates · distribution and follow-up.

Trainer debrief

Push hard on three things. First, **the override rule**: a group that scored the copper exceedance as 'medium' and left it out of the significant list has missed the point — a breached legal limit is significant regardless of the arithmetic. Second, **root cause depth**: any group whose 5 Whys ends at 'the operator did not check' has stopped at a person; make them push to the process. Third, **the vacant EHS Manager is the connective tissue** — it explains the stale register, the missed capacity assessment and the uninspected storage area. The best answers name it as a root cause of the others rather than as a fifth unrelated risk.

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